4 CHAPTER FOUR: PROPOSED PROJECT

4.1 Introduction

The development of any software system requires a structured and systematic approach that ensures the final product meets both user needs and technical requirements. One of the most effective ways of achieving this is by organizing the project into distinct phases that guide the progression of work from the initial idea to the final deployment. Each phase plays a critical role in reducing risks, clarifying objectives, and ensuring that the project maintains focus throughout its lifecycle.

For the proposed campus navigation system at United States International University – Africa (USIU-A), the project has been divided into a series of phases that align with standard software development practices while also reflecting the unique requirements of this study. These phases provide a step-by-step pathway that begins with research and requirements analysis, advances through design and implementation, and concludes with testing, evaluation, deployment, and documentation.

The purpose of this chapter is to present these proposed phases in detail, highlighting the activities carried out in each stage, their significance, and the expected outcomes. By structuring the project into clear phases, it becomes possible to manage resources effectively, monitor progress, and ensure that the final system is robust, user-friendly, and sustainable.

4.2 Project Phases

The successful development of the USIU-A campus navigation system depends on a carefully structured process that divides the project into manageable phases. Each phase represents a critical stage in the system's lifecycle, ensuring that the final product is both technically sound and user-centered. The phases follow a logical sequence beginning with research, moving through requirements gathering, design, implementation, and testing, before concluding with evaluation, deployment, and documentation.

4.2.1 Phase 1: Research Requirements Gathering and System Design

i. Research

The first activity in this phase is research, which lays the foundation for the USIU Campus Navigation System project. It involves reviewing both global and regional campus navigation platforms to identify best practices, strengths, and existing limitations. Global systems such as MIT WhereIs and NUS Campus Map with NextBus provide insights into features like interactive maps, route generation, and transport integration. On a regional scale, platforms such as the University of Cape Town campus maps and University of Nairobi wayfinding maps reveal the challenges of limited interactivity, basic layouts, and lack of real-time functionality. In addition, primary research will be conducted through surveys and interviews with USIU students, staff, and visitors to capture the unique navigation challenges faced on campus, such as finding classrooms, offices, or service centers in a timely manner.

ii. Requirements gathering

The second activity is requirements gathering, which transforms user needs into actionable system specifications. The functional requirements include searching for specific buildings, generating walking routes, viewing building details, and allowing administrators to update facility information. Non-functional requirements will focus on usability, system responsiveness, accessibility, scalability, and security. These requirements will be carefully analyzed, prioritized, and documented in a Software Requirements Specification (SRS) to serve as a guide for development. This ensures that the system is aligned with the realities of USIU's environment and addresses current limitations in campus orientation and accessibility.

iii. Proposal

The third activity is the preparation of the project proposal, which consolidates the research findings and requirements into a structured plan. The proposal clearly defines the system objectives, scope, methodology, and expected outcomes. It also justifies the project by highlighting the gaps in existing solutions, particularly the absence of an interactive and real-time navigation system within USIU. By presenting how the system will address these gaps, the proposal ensures a shared understanding among stakeholders and provides a reference framework for monitoring progress throughout the project's lifecycle.

iv. System Architecture and Design

The final activity in this phase is system architecture and design, which defines the structural and logical blueprint of the USIU Campus Navigation System. This includes creating use case diagrams to represent user interactions, data flow diagrams (DFDs) to illustrate the flow of information, and an entity-relationship diagram (ERD) to design the database structure. The system architecture will also specify the integration of the frontend (mobile/web application)**,** the backend (API server)**,** and the database (relational database such as PostgreSQL or MySQL)**,** along with external services like Google Maps API for route generation. These design artifacts provide a roadmap for implementation, ensuring the system is technically sound, user-friendly, and aligned with project goals.

4.2.2 Phase Two: System Development / Implementation

i. System Frontend

The system frontend will be developed as a mobile and web application using React for modularity, responsiveness, and cross-platform compatibility. The frontend will provide students, staff, and visitors with a clean and intuitive user interface that emphasizes simplicity and ease of use. Key features of the frontend include the ability to search for buildings, lecture halls, offices, or facilities, view interactive maps, and generate step-by-step walking routes across campus. The design will follow a structured layout with accessible navigation menus, clear icons, and interactive map features to ensure users can engage with the system effectively without requiring technical expertise. Special attention will be placed on responsive design so that the system functions seamlessly on mobile phones, tablets, and desktops.

ii. System Backend

The system backend will be implemented using Node.js with Express, providing the core business logic and ensuring smooth communication between the frontend and the database. The backend will handle functions such **as** user authentication, map data management, search queries, and route generation. It will also provide RESTful APIs to enable real-time communication with the frontend, ensuring that user actions, such as searching for a location or requesting a route, are executed reliably and securely. Scalability will be prioritized in the backend architecture so that the system can support future growth, including potential integrations with shuttle bus schedules, campus event data, or third-party services.

iii. Database

The database will be developed using PostgreSQL**,** a relational database management system (RDBMS) suited for handling structured and spatial data. PostgreSQL will manage core entities such **as** buildings, facilities, classrooms, offices, user accounts, and navigation routes. The database design will follow an entity-relationship model (ERD) to ensure data integrity and minimize redundancy. Since the project involves location data, PostgreSQL's PostGIS extension may be leveraged to handle geospatial queries and mapping functionality. In addition, indexing and query optimization will be implemented to improve performance, particularly during complex searches and when generating shortest-path routes across campus.

iv. System Integration

The system integration process will ensure that the frontend, backend, and database operate together seamlessly. RESTful APIs will act as the communication bridge, allowing the frontend to retrieve map data, building details, and navigation routes from the backend. Integration testing will be conducted to verify that each system component interacts correctly, especially during critical operations such as searching for a building, displaying its location, and generating a walking path. Continuous integration practices will be applied to ensure that updates and changes to the system do not disrupt its overall functionality, maintaining stability throughout development.

v. Core Functionality Implementation

This phase will focus on implementing the core functionalities that define the USIU Campus Navigation System. These include interactive campus maps, building search functionality**, real-**time route generation, and facility details (e.g., offices, lecture halls, and student services). Additional features such a**s** user-friendly map markers, accessibility routes for individuals with mobility needs, and administrative dashboards for updating building or facility information will also be implemented. By prioritizing these features, the project ensures that the Minimum Viable Product (MVP) directly addresses USIU's navigation challenges, making it easier for students, staff, and visitors to orient themselves on campus. This foundation will also allow for future expansions, such a**s** integration with shuttle services, event locations, and emergency safety features**.**

4.2.3 Phase Three: Testing, Documentation, And Presentation

i. Software Testing and Evaluation

The first step in this phase will involve unit testing, which will focus on verifying the smallest individual components of the system, such as frontend search forms, backend API endpoints, and database queries. Each unit will be tested independently to confirm that it performs its intended function correctly. For instance, the building search form will be tested to ensure that it retrieves accurate building details, while the database will be tested to confirm that it stores and retrieves location data correctly. Unit testing at this stage will help identify and fix errors early, reducing the cost of resolving issues later in the development process.

The next activity will be integration testing, which evaluates the interactions between the frontend, backend, and database. This ensures that the various system components work together seamlessly. Key scenarios, such as a student searching for a lecture hall, retrieving its location, and generating a walking route, will be tested end-to-end. Integration testing will confirm that data flows smoothly across the system and that the navigation experience is stable and reliable under different conditions.

Following this, system testing will be performed to evaluate the platform as a whole. This will include testing functional requirements (such as searching for buildings, displaying map locations, and generating walking paths) and non-functional requirements (such as response speed, usability, security, and scalability). The goal of this stage is to confirm that the USIU Campus Navigation System performs reliably under real-world conditions and aligns with the requirements set out in the Software Requirements Specification (SRS).

Finally, User Acceptance Testing (UAT**)** will be conducted to involve actual USIU students, staff, and visitors in evaluating the system. Test participants will be asked to perform tasks such as searching for specific classrooms, generating routes to offices, or locating facilities like the library or cafeteria. Their feedback will be collected and analyzed to refine usability, enhance the user experience, and validate that the system effectively addresses the navigation challenges it was designed to solve. By the end of this stage, the system will be validated against its objectives and prepared for deployment.

ii. Project Write-Up

All research findings, methodologies, system designs, implementation details, and testing results will be documented in a formal academic report. This write-up ensures that the USIU Campus Navigation System is well-documented for academic evaluation and also serves as a reference for potential future enhancements, scaling, or integration with other university systems such as transport services, event management, or emergency response systems.

iii. Final Presentation and Submission

The USIU Campus Navigation System will be presented to stakeholders, including lecturers, peers, and representatives of the student community. This final presentation will demonstrate the navigation challenges addressed, showcase the system's features such as interactive maps and route generation, and highlight its impact on improving orientation and accessibility at USIU. The demonstration will validate the project's success, confirm its contribution to solving real student challenges, and present opportunities for further expansion. The final report and system prototype will then be submitted for academic evaluation and assessment.

4.3 Program of work

The Program of Work defines the step-by-step execution of the project phases, ensuring that all objectives are met within the planned timelines. It serves as a structured framework for managing the project lifecycle, moving systematically from research and requirements gathering through to deployment. This organized approach supports efficient implementation, allows for early detection of possible challenges, and ensures optimal use of available resources. The following section presents a detailed breakdown of the work plan for each phase.

4.3.1 Phase One: Research, Requirements Gathering, And System Design

i. Research

Objective:

To understand navigation challenges at USIU and review existing global and regional campus navigation systems.

Tasks:

Review global (MIT WhereIs, NUS NextBus) and regional (UCT, UoN) systems;

conduct surveys and interviews with USIU students, staff, and visitors.

Deliverables:

Research findings report; identified gaps in current solutions.

Timeline: Week 1

ii. Requirements Gathering

Objective:

To define what the system should do.

Tasks:

Identify functional requirements (search, maps, route generation) and non-functional requirements (usability, scalability, accessibility).

Deliverables:

Software Requirements Specification (SRS).

Timeline: Week 2

iii. Proposal

Objective:

To consolidate research and requirements into a structured plan.

Tasks:

Draft abstract, background, literature review, aims & methodology tailored to USIU's needs.

Deliverables:

Completed and approved proposal.

Timeline:

Week 3

iv. System Architecture and Design

Objective:

To design the system structure and workflows.

Tasks:

Develop use case diagrams, ERDs, DFDs, wireframes, and overall architecture blueprint for the navigation system.

Deliverables:

System design blueprints and architecture document.

Timeline:

Week 4

4.3.2 Phase Two: System Development

i. Frontend Development

Objective:

To develop the user interface.

Tasks:

Build mobile/web app interfaces for students, staff, and administrators using React.

Deliverables:

Functional user interface with building search, interactive maps, and route display.

Timeline:

Weeks 5–6

ii. Backend Development

Objective:

To implement business logic and core services.

Tasks:

Develop backend APIs with Node.js/Express to handle authentication, map data management, and route generation.

Deliverables:

Functional backend with secure APIs and navigation logic.

Timeline:

Week 7

iii. API Integration

Objective:

To connect system components.

Tasks:

Link frontend, backend, and PostgreSQL database (with PostGIS for spatial data) via REST APIs.

Deliverables:

Fully integrated system with accurate data flow and route computation.

Timeline:

Week 8

iv. Deployment

Objective:

To make the system accessible.

Tasks:

Deploy the MVP of the Campus Navigation System on a cloud hosting service (e.g., AWS, Azure, or Heroku).

Deliverables:

Hosted live application accessible to users.

Timeline:

Week 9

4.3.3 Phase Three: Testing, Documentation, And Reporting

i. Software Testing and Evaluation

Objective:

To validate system functionality and performance.

Tasks:

Perform unit testing (frontend, backend, database), integration testing (end-to-end routes), and user acceptance testing with USIU students and staff.

Deliverables:

Testing report;

validated navigation system ready for release.

Timeline:

Weeks 10–11

ii. Project Write-up

Objective:

To document project outcomes.

Tasks:

Prepare final report including research, methodology, system design, implementation, testing, and results.

Deliverables:

Draft of final project report.

Timeline:

Week 12

iii. Final Presentation and Submission

Objective:

To present findings and deliver final work.

Tasks:

Conduct live demo of the USIU Campus Navigation System; prepare presentation slides; submit final project report.

Deliverables:

Final report and presentation slides.

Timeline:

Week 13

4.4 Ganntt Chart

Tasks within a project can be planned, scheduled, and their progress monitored using a Gantt chart, a visual project management tool. The Gantt chart functions as a comprehensive roadmap for the Asset Recovery System, outlining every stage of the project lifecycle, from planning and research to development, testing, and final presentation. By dividing the project into three primary stages, the chart facilitates efficient time management , defining task dependencies, setting start and finish dates, and emphasising important milestones.

The project will remain on course and all deliverables will be finished within the allotted time thanks to this well-organized timeline.

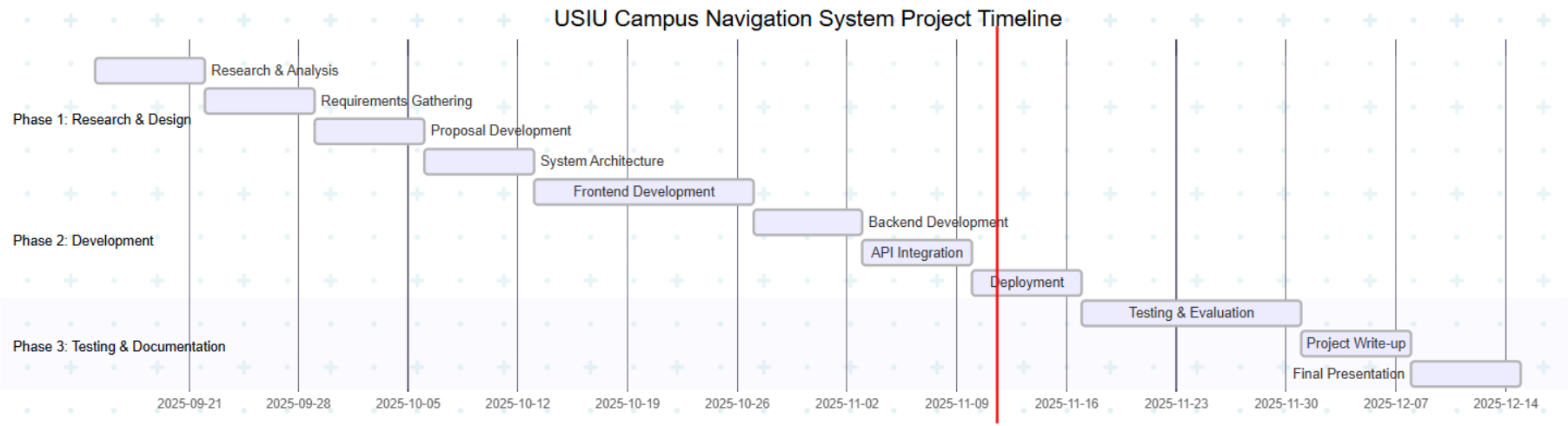

Figure 11: Gantt chart

Figure 11 above shows the Gantt chart that also highlights task dependencies and overlaps, providing a visual understanding of how activities are interrelated. This helps with effective project management by identifying critical paths and resource allocation needs. By following this structured plan, the development team can stay organized, minimize delays, and deliver the system within the projected timeframe. Moreover, the Gantt chart serves as a valuable tool for tracking progress and facilitating communication among team members. It enables regular updates and adjustments to the project schedule, ensuring that any risks or delays are identified early and addressed promptly. This proactive planning approach contributes to the successful execution of the Campus Navigation System by keeping the project aligned with its goals, deadlines, and stakeholder expectations.

Project Phases and Tasks

1. Phase One: Research and Requirements Gathering

This initial phase established the foundation of the Campus Navigation System by focusing on research, requirements gathering, proposal development, and system design. The tasks in this phase followed a structured sequence to ensure that the project was based on accurate data and user needs. The process began with comprehensive research to understand existing campus navigation solutions and identify gaps in accessibility, usability, and real-time location tracking. Following this, requirements gathering was carried out to define the system's functional and non-functional needs based on student, staff, and visitor feedback. Proposal development then formalized the system's objectives, scope, and implementation strategy. Finally, system design was undertaken, where the architecture, data flow, and navigation logic were defined. The successful completion of these tasks led to Milestone 1: Research and System Design Completed, marking the project's readiness to move into the development stage.

2. Phase Two: System Implementation

This phase concentrated on building and integrating the core components of the Campus Navigation System. It began with frontend development, where the user interface and user experience were created to allow smooth map navigation and intuitive access to building details. This was followed by backend development, which established the server-side logic, handled database connections, and supported real-time location queries. Next, API integration was performed to connect the system with mapping services such as Google Maps API and indoor positioning tools. Once the core functionalities were in place, deployment was done to host the system in a live environment, making it accessible for testing and feedback. The completion of this stage marked Milestone 2: System Implementation Completed, signifying that the Campus Navigation System was functional and ready for evaluation.

3. Phase Three: Testing, Documentation, and Presentation

The final phase emphasized ensuring system reliability, preparing documentation, and presenting the project outcomes. It began with software testing and evaluation, where the system was assessed for accuracy in navigation, performance, and user experience. Any detected bugs or inconsistencies were fixed to ensure stable operation. Following testing, project documentation was completed, summarizing all development processes, tools used, challenges encountered, and how they were addressed. The phase concluded with the final presentation and report submission, where the system was demonstrated to stakeholders and evaluators. This marked Milestone 3: Testing, Documentation, and Presentation Completed, confirming that the Campus Navigation System met its objectives and was ready for deployment within the university environment.

Dependencies

The project followed a logical flow of dependencies that ensured each task was completed before the next began. For instance, system design could only start after requirements gathering was finalized. Similarly, backend development depended on the completion of frontend structures, while API integration required both frontend and backend to be in place. Testing and documentation were dependent on the successful deployment of the system. These dependencies ensured a smooth transition between activities, reduced bottlenecks, and maintained consistent project progress.

Milestones

The project's milestones represented significant checkpoints that helped track progress and ensure the timely completion of critical stages. Milestone 1 marked the end of the research and system design phase, providing a validated blueprint for development. Milestone 2 represented the completion of system implementation, with all core functionalities successfully built and integrated. Finally, Milestone 3 indicated the completion of testing, documentation, and presentation, confirming that the system was fully functional, tested, and ready for use. These milestones provided structure, accountability, and clear progress tracking throughout the development of the Campus Navigation System.

4.5 Requirements

4.5.1 Hardware Requirements

The USIU Campus Navigation System will require both development and deployment hardware.

Development Machines

- Processor: Intel Core i5 (8th Gen or higher) or equivalent
- RAM: 8 GB minimum
- Storage: 256 GB SSD or higher
- Operating System: Windows 10/11, macOS, or Linux

End-User Devices

- Smartphones (Android/iOS): Users will primarily access the navigation system through mobile devices.
- Computers/Laptops: Access through a modern web browser (Chrome, Edge, Safari, or Firefox).
- Internet Access: Required for accessing maps, generating routes, and retrieving updated campus information.

4.5.2 Software Requirements

The following software tools and platforms will be used:

- Frontend: React, HTML5, CSS3, JavaScript
- Backend: Node.js with Express framework
- Database: PostgreSQL (with PostGIS extension for geospatial data)
- Version Control: Git & GitHub for source code management
- Development Tools: Visual Studio Code (IDE), Postman (API testing), pgAdmin (database management), Mapbox/Google Maps API (for mapping and route generation)
- Project Management Tools: MS Project/Project Plan 365 for task tracking, Canva/Figma for system diagrams and wireframes

4.5.3 Server Requirements

For hosting and deployment of the Campus Navigation System:

- Web Server: Node.js runtime environment hosted on a cloud platform (e.g., AWS, Heroku, or Render).
- Database Server: PostgreSQL with PostGIS, hosted on the same cloud service or as a managed PostgreSQL instance.
- Scalability Considerations: The server environment should allow for scaling to accommodate an increasing number of users (students, staff, and visitors) without compromising performance.

4.5.4 Budget Requirements

The budget for the USIU Campus Navigation System is minimal since most of the development tools and frameworks are open-source. However, some costs may be incurred for:

- Cloud hosting services (e.g., AWS, Heroku, or Render subscription)
- Domain name registration for a custom web address

- Third-party API services (e.g., Google Maps API or Mapbox, if usage exceeds the free tier)

4.6 Conclusion

This chapter has presented the methodology to be used in the design and development of the USIU Campus Navigation System, an interactive platform designed to simplify movement across campus. The work has been structured into three phases: Research and Design, System Development, and Testing & Reporting, each with well-defined objectives, tasks, and deliverables. The program of work and corresponding Gantt chart illustrate the timeline, dependencies, and milestones, ensuring that the project progresses in a systematic, efficient, and manageable way.